NPTEL Online Certification Courses

Indian Institute of Technology Kharagpur

Course Name: ETHICAL HACKING

Assignment- Week 6

TYPE OF QUESTION: MCQ/MSQ/SA

Number of questions: 10 **Total mark: 10 x 1 = 10**

QUESTION 1:

Which of the following attack types involves an attacker pretending to be a legitimate entity to gain unauthorized access?

a. Replay attack
b. Masquerade attack
c. Modification attack
d. Denial of service attack
e. Traffic analysis

Correct Answer: b

Detailed Solution: In a masquerade attack, an attacker pretends to be a legitimate entity or user to gain unauthorized access to a system or resource. Replay involves capturing and resending a transaction, modification involves altering a message in transit, DoS prevents legitimate access, and traffic analysis is a passive attack analyzing communication patterns. Thus, the correct option is (b).

QUESTION 2:

Encrypt the plaintext “SECURE” using a substitution cipher where each letter is replaced by its k-th next letter, assuming alphabets wrap around (Z followed by A) and k = 3.

a. VHFXUH
b. VFEXRU
c. UHFXVH
d. VHFXRV
e. None of these.

Correct Answer: a

Detailed Solution: Shifting each letter by 3: S→V, E→H, C→F, U→X, R→U, E→H. This gives the ciphertext “VHFXUH”.
Thus, the correct option is (a).

QUESTION 3:

Consider a mono-alphabetic cipher with the key mapping: (A B C D E F G H I J K L M N O P Q R S T U V W X Y Z) mapped to (Q W E R T Y U I O P A S D F G H J K L Z X C V B N M). What will be the encrypted form of the message “CODE”?

a. EGRT
b. EGDT
c. EODT
d. EGDR
e. None of these

Correct Answer: a

Detailed Solution: Using the given mapping: C→E, O→G, D→R, E→T. Hence, "CODE" encrypts to “EGRT”.
Thus, the correct option is (a).

QUESTION 4:

Which of the following statement(s) is/are **false** regarding hashing algorithms used for message integrity?

a. A good hash function should be resistant to collisions.
b. The same input will always produce the same hash output.
c. A hash function can be reversed to retrieve the original message.
d. Even a small change in input should produce a significantly different hash output.

Correct Answer: c

Detailed Solution: A good cryptographic hash function is collision-resistant, deterministic (the same input gives the same output), and exhibits the avalanche effect (small input changes drastically alter the output). Hash functions are one-way functions and cannot be reversed to retrieve the original input.

Thus, the correct (false) option is (c).

QUESTION 5:

Decrypt the ciphertext "KHOOR" which was encrypted using a substitution cipher with a shift of k = 3 (each letter replaced by its k-th next letter during encryption, the alphabets wrap around).

a. HELLO
b. WORLD
c. HELPS
d. HFLLO
e. None of these

Correct Answer: a

Detailed Solution: To decrypt, shift each letter back by 3: K→H, H→E, O→L, O→L, R→O. This gives "HELLO".

Thus, the correct option is (a).

QUESTION 6:

A DoS attack that makes a system or service inaccessible to legitimate users falls under which category of security attack?

a. Interruption
b. Interception
c. Modification
d. Fabrication
e. Replay

Correct Answer: a

Detailed Solution: A Denial-of-Service attack prevents legitimate users from accessing a system/service, which is classified as an interruption attack - an attack on availability.

Thus, the correct option is (a).

QUESTION 7:

For encryption using public-key cryptography, if sender A wants to send a confidential message to receiver B, which key is used to encrypt the message?

a. A's private key
b. A's public key
c. B's public key
d. B's private key

Correct Answer: c

Detailed Solution: In public-key cryptography, the sender encrypts the message using the receiver's public key, so that only the receiver can decrypt it using their corresponding private key.

Thus, the correct option is (c).

QUESTION 8:

_____ is the block size (in bits) used for plaintext blocks in the Data Encryption Standard (DES) algorithm.

Correct Answer: 64

Detailed Solution: DES processes plaintext in fixed 64-bit blocks, using a 56-bit key (with 8 additional parity bits, making the stored key 64 bits, though only 56 bits are effectively used for encryption).

Thus, the correct answer is 64.

QUESTION 9:

150 parties want to exchange messages securely using a public key encryption technique like RSA. How many distinct key values will be required?

Correct Answer: 300

Detailed Solution: In asymmetric (public-key) encryption, each party holds two keys (public and private). For N parties, total keys = 2N. For N = 150, 2 x 150 = 300.

Thus, the correct answer is 300.

QUESTION 10:

40 parties want to exchange messages securely using a symmetric key encryption algorithm. How many distinct keys will be required?

Correct Answer: 780

Detailed Solution: In symmetric encryption, every pair of communicating parties needs a separate key. For N parties, the number of keys = NC2 = N(N-1)/2. For N = 40, 40 x 39 / 2 = 780.
Thus, the correct answer is 780.

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